

Project Navi

Bachelor Designer & Developer

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PROJECT NAVI

Risks Analysis

Risk Identification

Sources of risk identification

The risks associated with this project were identified throughout the project planning phase using several project management tools and analyses. These tools helped anticipate potential issues that could affect the project's quality, cost, schedule, or user safety.

The main sources used to identify the risks were:

- **Project Scope**, to determine the technical and functional boundaries of the application.
- **SWOT Analysis**, to identify internal weaknesses and external threats.
- **Stakeholder Analysis**, to understand the expectations and constraints of the different actors involved in the project.
- **Feasibility Study**, to evaluate the project's technical, financial, operational, and legal viability.
- **Project Scenarios**, particularly the pessimistic scenario, to anticipate situations that could delay or compromise the project's success.

The identified risks were then classified into several categories to facilitate their analysis and management.

Risk categorization

Technical Risks

These risks concern the technologies used to develop the application.

Examples include:

- GPS inaccuracy.
- AI image recognition errors.
- Voice recognition failures.
- API service outages.
- Performance issues on older smartphones.

Functional Risks

These risks relate to the application's features and their ability to meet user needs.

Examples include:

- Features not meeting visually impaired users' expectations.
- Incorrect obstacle detection.
- Navigation instructions that are unclear or misleading.

Organizational Risks

These risks may affect project planning and execution.

Examples include:

- Development delays.
- Lack of user testing.
- Changes in project requirements during development.

Legal and Regulatory Risks

These risks concern compliance with applicable laws and standards.

Examples include:

- Non-compliance with GDPR.
- Insufficient accessibility compliance.
- Improper management of personal location data.

Security Risks

These risks involve protecting users and their personal information.

Examples include:

- Data breaches.
- Unauthorized access to user accounts.
- Loss of sensitive location information.

Financial Risks

These risks may impact the project budget.

Examples include:

- Unexpected API costs.
- Additional development expenses.
- Increased cloud hosting costs.

Safety Risks

As the application assists visually impaired users in real-world navigation, safety risks are particularly important.

Examples include:

- Failure to detect an obstacle.

- Delayed audio guidance.
- Incorrect navigation instructions leading to dangerous situations.

Conclusion

At this stage, the identified risks provide a comprehensive overview of the potential challenges that may affect the project's success. The next step consists of assessing each risk according to its probability and severity in order to prioritize mitigation actions.

Risk Assessment

Each identified risk was analyzed to understand its origin, triggering factors, potential consequences, and the stakeholders affected. This assessment provides a clearer understanding of the project's vulnerabilities before evaluating the probability and severity of each risk.

Risk	Origin	Trigger	Potential Consequences	Stakeholder
GPS inaccuracy	GPS limitations and environmental conditions	Dense urban areas, tunnels...	Incorrect navigation, user confusion, reduced reliability	End users, developer
Obstacle detection failure	AI recognition limitations	Poor lighting, weather conditions,	Missed obstacles, increased risk of accidents	End users

		unusual obstacles		
Voice recognition errors	Speech recognition technology	Background noise, unclear pronunciation	Incorrect commands, poor user experience	End users
API service outage	Third-party service dependency	Internet failure, API downtime, service interruption	Features become unavailable, degraded application performance	End users, developer
Development delays	Project management constraints	Underestimated workload, unexpected technical issues	Delayed delivery, increased development costs	Developer, project stakeholders
GDPR non-compliance	Poor data management	Inadequate handling of personal information	Legal penalties, loss of user trust	Developer, users
Data breach	Cybersecurity vulnerabilities	Weak authentication, software vulnerabilities	Exposure of personal and location data	Users, developer
Budget overrun	Financial estimation errors	Higher API or hosting costs	Budget increase, project scope reduction	Project owner
Accessibility issues	Inadequate interface design	Insufficient testing with visually	Poor usability,	End users

		impaired users	reduced adoption	
Delayed audio guidance	Performance limitations	High processing time, device limitations	Late instructions, increased safety risks	End users

Conclusion

This assessment highlights the main characteristics of each identified risk and their potential impact on the project. The next step is to evaluate each risk according to its **probability** and **severity** in order to determine its **criticality** and establish mitigation priorities.

Risk Evaluation and Prioritization

After identifying and assessing the project risks, each risk was evaluated according to its severity and probability. The objective of this evaluation is to determine the risk criticality, allowing the project team to prioritize mitigation actions and allocate resources efficiently.

The following scales were used throughout the analysis.

Severity Scale

Level	Description
1 – Negligible	Minimal impact on the project with no significant consequences
2 – Minor	Limited impact on quality, cost, or schedule.
3 – Moderate	Noticeable impact requiring corrective actions.
4 – Major	Significant impact on project objectives or user experience.
5 – Critical	Severe impact that could compromise user safety or the success of the project.

Probability Scale

Level	Description
1 – Very Low	Highly unlikely to occur.
2 – Low	Unlikely but possible.
3 – Medium	May occur during the project.
4 – High	Likely to occur.
5 – Very High	Expected to occur frequently or under normal conditions.

Risk criticality (Severity × Probability)

The criticality of each risk is calculated using the following formula:

Risk Criticality = Severity × Probability

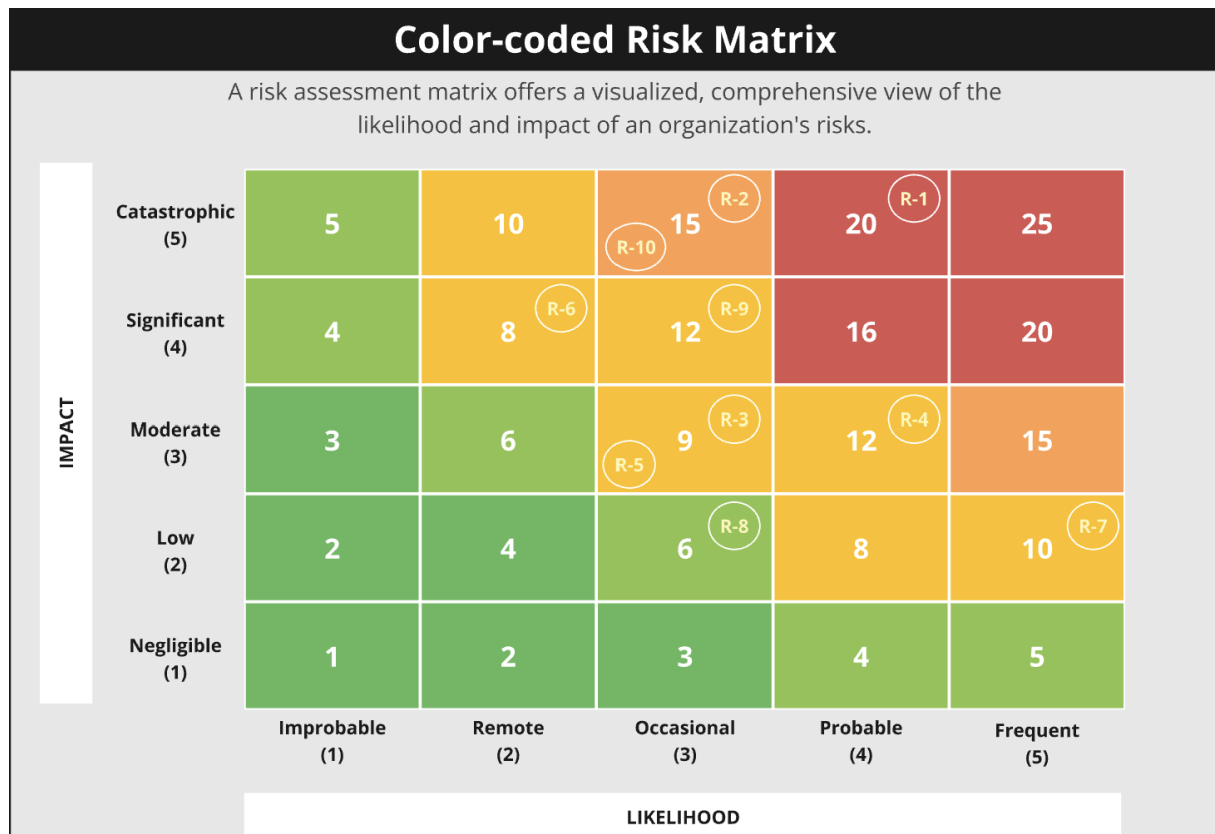
The higher the criticality score, the higher the priority assigned to the corresponding mitigation actions.

Risk	Severity	Probability	Criticality	Priority
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R1-GPS inaccuracy	5	4	20	Very High
R2-Obstacle detection failure	3	3	15	High
R3-Voice recognition errors	3	3	9	Medium
R4-API service outage	3	4	12	High
R5-Development delays	3	3	9	Medium
R6-GDPR non-compliance	4	2	8	Medium
R7-Data breach	5	2	10	High
R8-Budget overrun	3	2	6	Low
R9-Accessibility issues	4	3	12	High
R10-Delayed audio guidance	5	3	15	High

Risk matrix

The identified risks are represented by their corresponding IDs (R1–R10) according to their severity and probability scores.



Risk Mitigation Plan

After evaluating the identified risks, appropriate mitigation measures were defined to reduce their probability of occurrence and minimize their impact on the project. Each risk has been assigned to a responsible stakeholder, along with preventive and corrective actions to ensure effective risk management throughout the project lifecycle.

Risk	Criticality	Responsible	Prev. Actions	Corec. Action
GPS inaccuracy	20	Developer	Test multiple GPS providers, optimize positioning algorithms, perform field testing in different environments.	Recalculate the route automatically and notify the user when GPS accuracy is low.
Obstacle detection failure	15	Developer	Train and validate the AI model using diverse datasets, conduct extensive real-world testing.	Allow the user to retry the scan and continuously improve the detection model through updates.
Voice recognition errors	9	Developer	Support multiple speech patterns and test the application in noisy environments.	Ask the user to repeat the command or provide an alternative interaction method.
API service outage	12	Developer	Monitor API availability and implement fallback services where possible.	Switch to a backup service or activate an offline mode with limited functionality.

Development delays	9	Project Manager	Define realistic milestones, monitor progress regularly, and prioritize critical tasks.	Adjust the project schedule and reallocate resources if necessary.
GDPR non-compliance	8	Developer	Apply GDPR principles from the design phase, encrypt sensitive data, and collect only necessary information.	Update the application to meet legal requirements and inform affected users if required.
Data breach	10	Developer	Implement secure authentication, encrypt stored and transmitted data, and perform regular security testing.	Investigate the incident, fix vulnerabilities, and notify users in accordance with legal obligations.
Budget overrun	6	Project Owner	Estimate costs carefully and monitor expenses throughout the project.	Reduce non-essential features or identify alternative

				funding if required.
Accessibility issues	12	Developer	Involve visually impaired users in usability testing and follow accessibility guidelines.	Improve the interface based on user feedback and release corrective updates.
Delayed audio guidance	15	Developer	Optimize processing time and minimize latency during navigation.	Prioritize audio processing and provide immediate warning messages whenever possible.

Conclusion

The mitigation plan ensures that each identified risk is associated with appropriate preventive and corrective measures. By assigning clear responsibilities and defining response strategies, the project team can reduce the overall impact of potential issues and improve the reliability and safety of the application for visually impaired users.

Risk Monitoring

Risk management is an ongoing process throughout the project lifecycle. Regular monitoring ensures that identified risks remain under control and that new risks are detected as the project evolves.

To achieve this, the following monitoring activities will be carried out:

- Regularly review the risk register to verify the status of each identified risk.
- Reassess the severity and probability of risks whenever significant changes occur in the project.
- Monitor the effectiveness of preventive and corrective actions.
- Identify and document any new risks that emerge during development or testing.
- Update the mitigation plan whenever necessary to reflect changes in the project's scope, technology, or regulatory requirements.

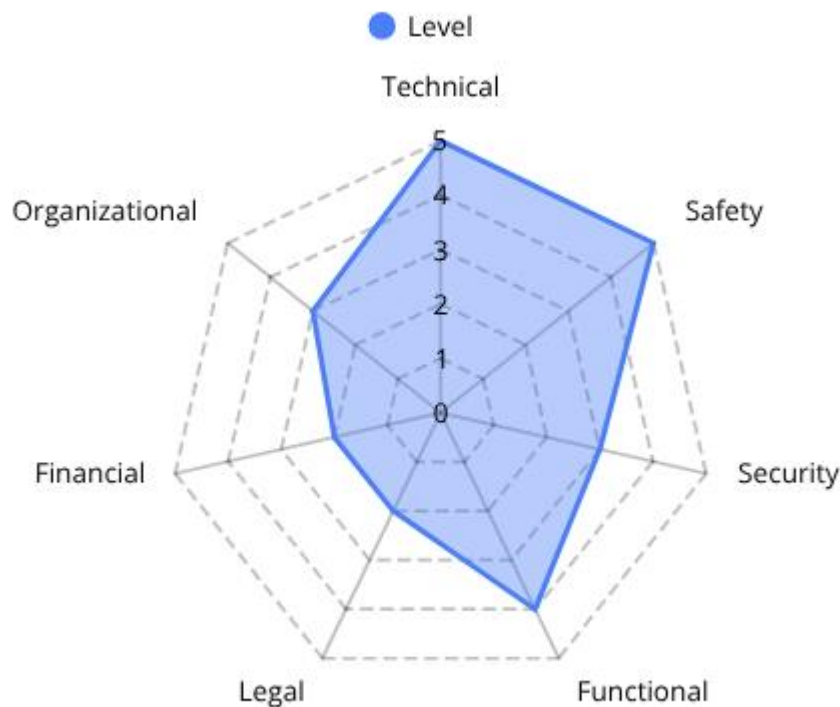
Progress will be reviewed at each major project milestone, including the completion of key development phases and user testing sessions. Risks classified as **High** or **Very High** will receive particular attention to ensure that mitigation measures remain effective.

Conclusion

Continuous risk monitoring helps maintain the project's stability and improves decision-making throughout the development process. By regularly reviewing and updating the risk management plan, the project team can respond proactively to potential issues and ensure the delivery of a reliable and safe application for visually impaired users.

Risk Analysis Summary

Radar chart



The radar chart provides an overall view of the project's exposure across the main risk categories. The analysis shows that Technical and Safety risks have the highest impact due to the application's reliance on GPS, artificial intelligence, and real-time navigation. Functional risks are also significant, as the application must accurately meet the needs of visually impaired users. Security and Organizational risks remain moderate, while Legal and Financial risks are considered lower but continue to be monitored throughout the project.

Overall analysis and conclusions

The risk analysis conducted for this project has identified the main threats that could affect the successful development and deployment of the application. By evaluating each risk

according to its severity and probability, it was possible to prioritize the most critical issues and define appropriate mitigation strategies.

The analysis highlights that the most significant risks are directly related to user safety, particularly GPS inaccuracies, obstacle detection failures, and delayed audio guidance. These risks received the highest criticality scores because they could compromise the application's primary objective: providing reliable assistance to visually impaired users.